

Syntax of a for loop in Python

Here's the basic structure of the loop body:

for **loop_variable** in **sequence**:

Code block to execute

Component	Explanation
for	This keyword starts the for loop.
A loop variable	This Python loop variable takes the value of each item in the sequence, one at a time. You can name it anything you like. In our example structure, it's loop_variable .
in	This keyword is used to specify the sequence that you want to iterate over.
A sequence	This is the collection of items you want to loop through. In our example structure, it's sequence .

Let's say you have a list of numbers, and you want to print each number.

```
numbers = [1, 2, 3, 4, 5]
```

```
for number in numbers:
```

```
    print(number)
```

In this loop example, the following functions and variables make this operation work:

- numbers is the sequence, which in this case is a defined list of numbers.
- number is the loop variable that takes each value from the list, one at a time.
- print(number) is a statement in the code block that gets executed for each item in the list.

This loop goes through each number in the list and prints it to be the following:

```
1
2
3
4
5
```

How to use a for loop in Python

Let's go through the mechanics of for loops in different use cases to better understand their potential applications.

